# Same Motion, Different Scores: Supplementary Material

Anonymous CVPR submission

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This supplement contains the equivalence proof and verification (Sec. S1), equal-budget and physical controls (Secs. S2 and S3), the full public-output audit (Sec. S4), attack protocols (Sec. S5), the sign-language experiments (Sec. S6) and statistical details (Sec. S7). Code, configurations, saved outputs and checkpoint hashes accompany the submission.

## S1. Equivalence and verification

**Fiber.** For $a \neq 0$ and $b$ not parallel to $a$, let $u = a/\|a\|$, $b_\perp = b - (u^\top b)u$ and $v = b_\perp/\|b_\perp\|$. Under $a' = \alpha a$, $b' = \beta b + \eta a$ with $\alpha, \beta > 0$, the first vector normalizes to $u$ and $b'_\perp = \beta b_\perp$, so $v$ and $R = [u; v; u \times v]$ are unchanged. Conversely, every $d$ with representative $(u, v)$ is $a = \rho u$, $b = \sigma v + \tau u$ with $\rho, \sigma > 0$, and any two such encodings are related by one recoding. Recodings compose as $(\alpha_2\alpha_1,\ \beta_2\beta_1,\ \beta_2\eta_1 + \eta_2\alpha_1)$, with inverse $(\alpha^{-1}, \beta^{-1}, -\eta/(\alpha\beta))$. Negative scales can flip rows and are excluded.

**Canonicalization.** $\mathcal{C}(a, b) = (u, v)$ is idempotent and constant on fibers, so $f \circ \mathcal{C}$ is invariant for any deterministic reader $f$. Its derivative vanishes along $(a, 0)$, $(0, b)$ and $(0, a)$, the three fiber directions.

**Numerical checks.** On 4,096 random double-precision features, fiber reconstruction error is $4.6 \times 10^{-15}$, canonical invariance error $4.9 \times 10^{-15}$ and directional derivatives are below $3 \times 10^{-14}$; our converter matches the official one to $4.5 \times 10^{-16}$. For every experiment, saved arrays are checked independently of any score: the maximum rotation discrepancy is below $4 \times 10^{-5}$ degrees, HumanML3D position channels are bitwise unchanged, sampled forward kinematics differs by at most $9.6 \times 10^{-7}$, and all non-rotation channels and padding are bitwise identical. Because tiny floating-point differences can flip a near-tied decoder, hypothesis equality is verified on saved strings rather than inferred.

Table S1. **Equal-budget controls, $\kappa = 3$, mean of 20 draws.** Left: T2M $\Delta$FID; right: TMR $\Delta$R1 in percentage points. Every fixed-fiber T2M interval excludes zero; all intervals and $\kappa = 1$ are in the released tables.

| | T2M $\Delta$FID | | | TMR $\Delta$R1 (pp) | | |
|---|---|---|---|---|---|---|
| Block | random | fixed + | fixed − | random | fixed + | fixed − |
| Fiber | 0.0024 | 0.0823 | 0.1169 | +0.01 | −0.10 | −0.23 |
| Rotation | 0.0035 | 0.0106 | 0.0151 | +0.02 | −0.22 | +0.01 |
| Velocity | 0.0159 | 0.0603 | 0.4760 | −0.00 | −0.22 | −0.19 |
| Contact | 0 | 0 | 0 | −0.24 | −0.84 | −2.32 |
| Position | 0.0183 | 0.3416 | 1.1890 | +0.03 | −0.18 | −0.08 |

## S2. Equal-budget block controls

For evaluator $r$ with fixed per-channel standard deviations $D_r$, each intervention on a clip of $L$ valid frames is rescaled to $\|D_r^{-1}(y - x)\|_2 = \kappa\sqrt{L}$, counting all 263 channels. *Fiber* interventions change only redundant coordinates; *rotation* interventions apply a local SO(3) change and re-encode it canonically; the other blocks receive additive changes to velocities, contacts or positions only. *Random* uses i.i.d. Gaussian directions; *fixed* uses one direction shared by all joints and frames, with both signs. Only fiber interventions keep every represented quantity fixed. The T2M evaluator does not read contacts.

Within the rotation block, the two fixed fiber directions change T2M FID 7.76× and 7.73× more than the fixed rotation directions of the same sign; at $\kappa = 1$ the ratios are 8.07 and 7.68. Fixed directions in non-rotation blocks can have larger effects (Tab. S1). What distinguishes the fiber is that it changes no represented quantity, so no response to it is warranted.

## S3. Physical calibration

We rotate the 21 non-root joints of real HumanML3D motion by a local angle about a random axis, either fixed through the clip or redrawn per frame, and recompute rotations, positions, velocities and contacts with the native skeleton. All conditions are compared with the same zero-

Table S2. **FID of physical and encoding changes** against the same reconstructed real motion, mean of 20 draws.

| Angle | 1° | 3° | 10° | 30° |
|---|---|---|---|---|
| Fixed axis | 0.0001 | 0.0026 | 0.300 | 14.44 |
| Per-frame axis | 0.350 | 10.24 | 38.98 | 75.17 |

Recodings (no physical change): fixed ±0.1: 0.059 / 0.083; fixed ±0.35: 1.97 / 3.57; random 0.35: 0.024.

angle reconstruction, whose own FID to the original data is 0.020 (R1 0.451→0.447).

## S4. Public-output audit

**Protocol.** We use the native HumanML3D test loader with its caption, crop and mirroring draws (4,640 entries after dropping the last incomplete batch of 32) and 20 seeds; the real reference is an independent draw. MDM, MotionDiffuse and DiP emit 1,024 outputs per draw and the others 4,640; KIT-ML uses 704. Native inference settings are kept: MDM 50 steps (CFG 2.5), MLD DDIM-50 (CFG 7.5), MotionLCM one step, T2M-GPT categorical sampling, T2M with its length estimator, MoMask 18 mask steps, MotionDiffuse 1,000-step DDPM, and DiP with a 20-frame prefix excluded from scoring. Raw and canonical scores share every output, caption, candidate batch and reference. Intervals are paired $t_{19}$, Bonferroni-adjusted over ten FID/R1 endpoints per study.

**Second checkpoint and retrieval.** Table S3 repeats the original five releases with the distinct official T2M weights distributed with T2M-GPT. Four FID gaps exclude zero there; MLD's is inconclusive and T2M's changes sign. Retrieval gaps are small with both checkpoints and with TMR (Tab. S4).

**Sign of the gap and model family.** Figure S1 plots each release's gap against how far its raw rotations are from canonical. Three of four diffusion or consistency releases have negative gaps and both VQ releases positive ones; the distance itself does not predict the gap (Spearman $-0.10$ signed and $0.29$ absolute; exact permutation $p = 0.84$ and $0.50$). Removing rotation non-canonicality therefore does not uniformly favor any model family.

**Essential-feature readers.** The public MARDM GRU and contrastive readers, which read only the 67 root and position features, are exactly invariant to rotation recoding. Their release rankings differ from that of canonical T2M by two adjacent swaps each (Kendall $\tau = 0.857$ for both).

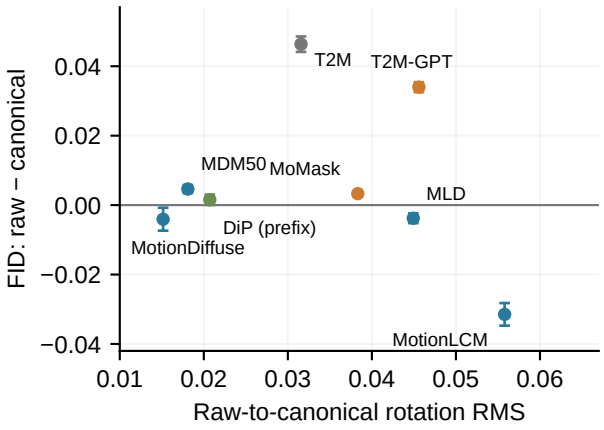



Figure S1. **Gap versus distance from canonical rotations** for the eight HumanML3D releases, with paired intervals. Blue: diffusion/consistency; orange: VQ; gray: VAE; green: prefix-conditioned.

**Published margins.** Table 1 of the main paper compares gaps with author-reported 95% margins. These margins describe different protocols and are used only for scale. We use MDM's corrected 0.489±0.047, T2M with estimated lengths, MoMask's final KIT table and MotionDiffuse's release preprint.

## S5. Attack protocols

**Targeted recoding.** We sample 1,024 full-caption test clips (40–199 frames) once, with fixed captions and crops, and use them for real motion, MDM outputs and both evaluators. Per-frame, per-joint coefficients start at zero and are optimized for 80 Adam steps (learning rate 0.1, bound 0.35) to minimize embedding distance to the actual or a circularly shifted caption, with evaluator weights frozen. R-Precision averages five fixed 32-candidate permutations. Optimizing through a canonicalizing evaluator leaves every score at its baseline. For T2M, the actual-caption attack worsens FID (MDM 0.37→1.29): it moves retrieval, not every metric.

**Caption-blind recoding.** For 960 fresh MDM outputs whose source clips are disjoint from the targeted study, we compute embedding moments $(\mu_s, \Sigma_s)$ and those of 2,048 training motions $(\mu_t, \Sigma_t)$, and set per-clip targets $\mu_t + (e_i - \mu_s)T$ with the Gaussian transport map [1]

$$T = \Sigma_s^{-1/2}\left(\Sigma_s^{1/2}\Sigma_t\Sigma_s^{1/2}\right)^{1/2}\Sigma_s^{-1/2}, \qquad (1)$$

with eigenvalues floored at $10^{-4}\,\mathrm{tr}(\Sigma)/512$. Coefficients minimize squared distance to these targets for 80 Adam steps. No caption content, test reference or test-time score enters the optimization, and all outputs are saved before ref-

Table S3. **Raw minus canonical with two official T2M checkpoints**, 20 draws. Intervals adjust ten endpoints per checkpoint.

| Checkpoint | Release | Raw FID | $\mathcal{C}$ FID | FID gap 95% CI | Raw R1 | $\mathcal{C}$ R1 | R1 gap 95% CI |
|---|---|---|---|---|---|---|---|
| MDM-linked | MDM | 0.4550 | 0.4504 | $[+0.0035, +0.0058]$ | 0.4115 | 0.4115 | $[-0.0017, +0.0018]$ |
| MDM-linked | MLD | 0.4453 | 0.4491 | $[-0.0052, -0.0024]$ | 0.4118 | 0.4105 | $[+0.0003, +0.0023]$ |
| MDM-linked | MotionLCM | 0.5747 | 0.6062 | $[-0.0347, -0.0282]$ | 0.4374 | 0.4350 | $[+0.0005, +0.0043]$ |
| MDM-linked | T2M-GPT | 0.1828 | 0.1488 | $[+0.0325, +0.0354]$ | 0.4226 | 0.4240 | $[-0.0030, +0.0002]$ |
| MDM-linked | T2M | 1.3935 | 1.3471 | $[+0.0441, +0.0486]$ | 0.3933 | 0.3930 | $[-0.0009, +0.0017]$ |
| T2M-GPT | MDM | 0.5129 | 0.5104 | $[+0.0012, +0.0040]$ | 0.4586 | 0.4584 | $[-0.0015, +0.0019]$ |
| T2M-GPT | MLD | 0.4206 | 0.4216 | $[-0.0023, +0.0002]$ | 0.4669 | 0.4666 | $[-0.0012, +0.0017]$ |
| T2M-GPT | MotionLCM | 0.4195 | 0.4294 | $[-0.0138, -0.0060]$ | 0.5042 | 0.5026 | $[-0.0004, +0.0035]$ |
| T2M-GPT | T2M-GPT | 0.1204 | 0.1115 | $[+0.0076, +0.0103]$ | 0.4909 | 0.4898 | $[-0.0004, +0.0026]$ |
| T2M-GPT | T2M | 1.0836 | 1.0917 | $[-0.0111, -0.0051]$ | 0.4547 | 0.4540 | $[-0.0007, +0.0021]$ |

Table S4. **TMR retrieval, raw minus canonical**, 20 draws; intervals adjust twelve endpoints.

| Release | Raw R1 | R1 gap 95% CI |
|---|---|---|
| MDM | 0.5578 | $[-0.0015, +0.0029]$ |
| MLD | 0.5848 | $[-0.0005, +0.0019]$ |
| MotionLCM | 0.6785 | $[+0.0013, +0.0043]$ |
| T2M-GPT | 0.6703 | $[-0.0006, +0.0017]$ |
| T2M | 0.5626 | $[+0.0003, +0.0028]$ |

Table S5. **Caption-blind FID**, 960 outputs per cohort. *Source*: evaluator whose gradients were used; *other*: the second official T2M checkpoint, which supplied no gradients. Canonical-gradient controls stay at the original value.

| Cohort | Evaluator | Original | $b = 0.1$ | $b = 0.35$ |
|---|---|---|---|---|
| MDM seed 42 | source | 0.384 | 0.210 | 1.085 |
| | other | 0.408 | 0.334 | 0.972 |
| MDM seed 4242 | source | 0.376 | 0.176 | 1.148 |
| | other | 0.371 | 0.303 | 0.856 |
| MDM seed 7777 | source | 0.327 | 0.199 | 1.429 |
| | other | 0.379 | 0.284 | 0.779 |
| MLD | source | 0.706 | 0.289 | 1.611 |
| | other | 0.659 | 0.579 | 1.345 |

erence scoring. Two bounds, 0.1 and 0.35, were fixed in advance; the larger raises FID in every cohort (Tab. S5).

## S6. Sign language

**Data and readers.** We use the released SignSparK body and hand rotation features (240 per frame: 10 body and $2 \times 15$ hand joints) and the public backtranslation readers, one Sign Language Transformer per corpus, with native tokenization, beam search and 300-frame cap. Full test sets contain 642 PHOENIX-2014T, 2,183 How2Sign and 1,162 CSL-Daily clips; real-motion BLEU is 15.00, 3.48 and 6.89. Left-hand channels, which the released pipeline preprocesses canonically, are unchanged in every attack.

**Learned layers.** The PHOENIX layer (299k parameters) reads a frozen text-to-motion diffusion model's output and the source sentence, encoded by the generator's frozen text encoder, and predicts bounded coefficients with cross-attention and three dilated temporal convolutions. It is trained for 20 epochs on 7,096 training motions with the reader's caption cross-entropy; generator and reader are frozen and final checkpoints are used. The How2Sign and CSL-Daily layers (770k parameters) read real training motion and the reader's own word embeddings. Table S6 reports ablations: without the source sentence the gain is small, and with the loss computed after canonicalization it vanishes, as Eq. (4) of the main paper predicts.

Table S6. **Learned-layer ablations**, full test sets. Raw BLEU and gain over no layer with paired 95% intervals; canonical hypotheses are unchanged in every row. How2Sign/CSL controls use one seed.

| Corpus | Layer input / loss | Raw | Gain [95% CI] |
|---|---|---|---|
| PHOENIX | text+motion / raw | 15.21 | 8.81 [7.82, 9.87] |
| | motion only / raw | 7.60 | 1.21 [0.66, 1.76] |
| | text+motion / canonical | 6.26 | $-0.13$ [$-0.32$, 0.04] |
| How2Sign | text+motion / raw | 6.11 | 2.64 [2.27, 3.03] |
| | text+motion / canonical | 3.48 | 0.00 [$-0.18$, 0.20] |
| CSL-Daily | text+motion / raw | 7.43 | 0.54 [0.14, 0.97] |
| | text+motion / canonical | 6.79 | $-0.10$ [$-0.40$, 0.09] |

**Per-clip attacks.** On 128 real test clips per corpus, 80 Adam steps toward a circularly shifted sentence raise BLEU against that sentence to 85.19 (PHOENIX), 47.34 (How2Sign) and 32.44 (CSL-Daily), while BLEU against the actual sentence drops to 0.00, 0.00 and 1.26; canonical hypotheses are unchanged. A gloss-level CTC recognizer of a different architecture shows the same failure (wrong-target WER 118%→4%).

**Detector statistics.** Write $s = (\rho - 1, \sigma - 1, \tau)$. The squared distance to the canonical encoding is $E = \|s\|^2$, and the squared feature norm satisfies $N - E = 2(\rho + \sigma) - 2$, so rotating $s$ about $(1, 1, 0)/\sqrt{2}$ preserves both. A per-clip attack restricted to this circle (200 steps) raises PHOENIX

BLEU from 6.62 to 10.72 (+4.09, 95% CI [3.04, 5.24]), while a threshold detector on the RMS distance to canonical has AUC 0.500. Fixing the full $2 \times 2$ Gram matrix of $(a, b)$ would leave no channel; matching a few statistics is therefore not a general defense, whereas canonicalization closes the whole fiber.

**Reader adaptation.** Real How2Sign and CSL-Daily features are not canonical. We fine-tune each public reader for five epochs on raw or on canonical training features, with three seeds each and matched batches. Test BLEU is 3.37 raw versus 3.47 canonical on How2Sign (difference $+0.10$, 95% CI $[-0.01, 0.21]$) and 7.24 versus 7.22 on CSL-Daily ($-0.02$, $[-0.19, 0.15]$). Canonical readers keep every hypothesis under fresh attacks and under fixed equivalent recodings of all 10,035 test inputs.

## S7. Statistics and reproducibility

Motion audits use paired $t_{19}$ intervals over 20 native draws, Bonferroni-adjusted within each stated family. Sign-language intervals use 20,000 paired bootstrap resamples that recompute corpus BLEU from sufficient statistics, clustered by the 149 official How2Sign video identifiers; seed sensitivity additionally resamples the three fitting seeds. Corpus BLEU and WER are recomputed with independent implementations. Clip selections, seeds and bounds were fixed before scoring, and no checkpoint was selected on test scores. Experiments ran with PyTorch 2.11 and CUDA 12.8 on one RTX PRO 6000; exact floating-point decoding can depend on hardware.

## References

[1] Rajendra Bhatia, Tanvi Jain, and Yongdo Lim. On the Bures–Wasserstein distance between positive definite matrices, 2017. 2